DGMS Examination 2019 — Question 54 Detailed Solution

Junior Scientific Officer (Geology) | Fluvial Geomorphology & Structural Control

QUESTION 54

Which one of the following pairs is NOT correctly matched?

(a) Annular drainages : Develop on eroded dome in alternate resistant/weak sediments

(b) Parallel drainages : Develop on horizontal sediments or homogeneous crystalline rocks
NOT CORRECTLY MATCHED (CORRECT OPTION)

(c) Trellis drainages : Develop on alternating weak and resistant tilted sedimentary rocks

(d) Discordant drainages : Develop on river channels cut across geological structures

1. Executive Summary & Answer Verification

The correct option to choose is **(b) Parallel drainages : Develop on horizontal sediments or homogeneous crystalline rocks**.

In fluvial geomorphology and structural geology (A. D. Howard, 1967; Thornbury, 1969; Arthur L. Bloom):

- On flat, horizontally bedded sedimentary strata or uniformly resistant, homogeneous crystalline rocks (e.g., massive granite or flat volcanic plateau basalts), streams branch organically and randomly in all directions like the limbs of an oak tree with acute junction angles. This is universally termed a **Dendritic drainage pattern** (from Greek *dendron* = tree).
- A **Parallel drainage pattern** develops strictly where there is a pronounced, uniform, steep regional slope or regional structural dip (such as steep monoclines, fault-scarp slopes, or narrow coastal plains), forcing both master streams and tributaries to flow parallel to one another down the regional gradient. It does *not* develop on flat horizontal or massive rocks.

Therefore, statement (b) erroneously assigns the diagnostic geological setting of a *dendritic* pattern to a *parallel* pattern, making it NOT correctly matched.

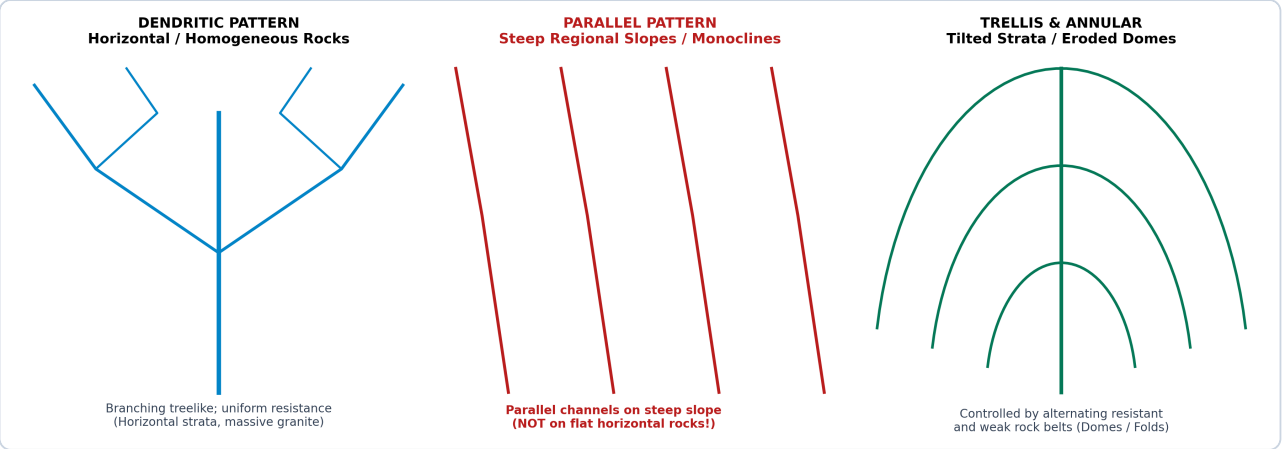


Figure 54.1: *Primary drainage patterns governed by structural geology: (Left)* ***Dendritic*** *develops on flat horizontal strata or homogeneous crystalline rocks; (Center)* ***Parallel*** *requires steep regional slopes/monoclines; (Right)* ***Trellis & Annular*** *develop on folded or domed strata with alternating resistant and weak rock layers.*

2. Detailed Geomorphological Analysis of Drainage Patterns

2.1 Dendritic vs. Parallel Patterns

1. **Dendritic Pattern (Tree-like):**
   - **Geological Setting:** Horizontal sedimentary beds or completely massive, isotropic crystalline igneous/metamorphic rocks without preferred fracture orientation.
   - **Mechanism:** Because resistance to erosion is isotropic (uniform in all directions), tributaries join larger streams at acute angles ($<90^\circ$) without any preferred structural alignment.
2. **Parallel Pattern:**
   - **Geological Setting:** Elongated landforms with steep, uniform regional slopes (e.g., coastal plains, steep volcanic cone flanks, or tilted homoclinal blocks).
   - **Mechanism:** Strong slope gradient overrides minor lithological differences, compelling all stream channels to run parallel or sub-parallel down the dominant slope direction.

2.2 Verification of the Other Drainage Classes

1. **(a) Annular Drainage — Correctly Matched:** A ring-like or concentric drainage network that develops on maturely dissected structural domes or structural basins. As concentric bands of alternating resistant (sandstone/limestone) and weak (shale) strata are exposed, streams excavate concentric strike valleys in the softer rock, draining along curved annular paths (e.g., the Black Hills of South Dakota).
2. **(c) Trellis Drainage — Correctly Matched:** A rectilinear pattern where major parallel trunk streams flow in longitudinal strike valleys underlain by weak strata, and short subsequent tributaries enter at right angles (90°) from adjacent ridges of resistant rock. Diagnostic of folded, tilted sedimentary belts (anticlines and synclines, such as the Appalachian Valley and Ridge province).
3. **(d) Discordant Drainage — Correctly Matched:** Refers to a river system that does not conform to the underlying geological structure or topography, cutting transverse gorges across resistant ridges. The two classical genetic mechanisms are:
   - *Antecedent Drainage:* The river established its course before the tectonic uplift of the ridge or mountain, and eroded its channel downward at a rate equal to or exceeding the uplift rate (e.g., the Brahmaputra, Indus, and Sutlej rivers cutting through the Himalayas).

- *Superimposed (Epigenetic) Drainage:* The river established its course on an overlying flat sedimentary cover; as erosion stripped the cover, the river incised blindly into the discordant structures of the buried basement.

3. Systematic Comparison of Drainage Patterns and Bedrock Controls

Drainage Pattern	Network Morphology	Controlling Geological Setting	Exam Option Status
Dendritic	Irregular branching tree-like; acute angles	**Horizontal strata or homogeneous crystalline rock**	**True setting of the description in (b)**
Parallel	Parallel, straight, swift stream channels	**Steep regional slopes, coastal monoclines, fault scarps**	**MISMATCHED in (b)**
Annular	Concentric ring-like or circular paths	Maturely eroded domes/basins in alternating weak/hard rock	Correctly matched in (a)
Trellis	Right-angled tributaries entering parallel trunks	Tilted/folded strata of alternating resistant and weak rock	Correctly matched in (c)
Discordant	Transverse gorges cutting across structural ridges	Antecedent or superimposed river incision across structures	Correctly matched in (d)
Rectangular	Sharp 90° bends and right-angled junctions	Orthogonal joint sets and intersecting fault systems	Standard structural type
Radial	Radiating outward like wheel spokes	Isolated volcanic cones, circular domal peaks	Standard structural type

Textbook Authority

Arthur D. Howard (1967), *Drainage Analysis in Geologic Interpretation: A Summation* (AAPG Bulletin):
"Dendritic: Characterized by irregular branching in all directions... indicates horizontal sediments or uniformly resistant, crystalline rocks.
Parallel: Consists of parallel or subparallel streams... caused by pronounced, uniformly steep regional slope or parallel topographic features."
William D. Thornbury, *Principles of Geomorphology*:
"Dendritic drainage patterns are characterized by treelike branching and develop on rocks of uniform resistance to erosion, such as flat-lying sedimentary beds or massive igneous rocks... Parallel drainage patterns develop on steep, unidirectional structural slopes."